LMS PLATFORM

Infrastructure & Cost Proposal

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EXECUTIVE SUMMARY

This document outlines the 3-phase infrastructure plan and associated third-party service costs for building an online Learning Management System (LMS). Development costs are excluded.

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Phase 1 — Service Cost Breakdown

Service	Free Tier	Monthly Cost	Notes
Cloudinary (Video)	25GB storage + 25GB bandwidth	**$0 / month**	Video hosting & delivery
MongoDB Atlas	512MB database	**$0 / month**	Database hosting
Vercel (Frontend)	Unlimited hobby	**$0 / month**	React frontend hosting
Render (Backend)	Free tier (sleeps)	**$0 – $7 / month**	$7/mo recommended to avoid sleep
SendGrid (Email)	100 emails/day	**$0 / month**	Transactional email
Domain Name	—	**~$1 / month**	~$12/year (optional for MVP)
PHASE 1 TOTAL		**$0 – $8 / month**	**Effectively FREE to launch**

Note: Cloudinary free tier supports approximately 17–35 hours of watched content per month. Video files should be limited to 720p during MVP to maximize free tier usage.

3. Phase 2 — Core LMS + Payment Integration

Scope

- Stripe payment integration (replace manual flow)
- Instructor revenue split & payouts via Stripe Connect
- Video streaming upgrade: AWS S3 + CloudFront OR Mux
- Course progress tracking per student
- Quizzes & completion certificates (PDF generation)
- Reviews and ratings system
- Coupon and discount code system
- Advanced email sequences (progress reminders, promotions)

Phase 2 — Service Cost Breakdown

Service	Plan	Monthly Cost	Notes
Mux (Video Streaming)	Pay per use	**~$20–$80 / month**	$0.015/min stored + $0.0025/min delivered. Scales with usage.
OR: AWS S3 + CloudFront	Pay per use	**~$15–$60 / month**	More control, more setup. Scales well.
Stripe Payments	Per transaction	**2.9% + $0.30 / txn**	No monthly fee. Deducted from revenue.
Stripe Connect (Payouts)	Per payout	**0.25% per payout**	For instructor revenue splitting
MongoDB Atlas	M10 Cluster	**$57 / month**	Upgrade needed for production scale
Render / Railway (Backend)	Starter / Pro	**$20–$25 / month**	Always-on server with more resources
SendGrid	Essentials	**$19.95 / month**	50,000 emails/month
Vercel (Frontend)	Pro	**$20 / month**	Teams, analytics & more bandwidth
PHASE 2 TOTAL (est.)		**$155–$220 / month**	**Excludes Stripe txn fees (from revenue)**

Important: Stripe transaction fees (2.9% + $0.30) are deducted from course sales revenue, not a fixed monthly overhead. At $10,000/month in sales, Stripe fees are approximately $320.

4. Phase 3 — Scale & Growth

Scope

- Mobile application (React Native — iOS & Android)
- Advanced search powered by Algolia or Elasticsearch
- Recommendation engine for personalized learning
- Admin content moderation panel
- Analytics dashboard for instructors & admins
- Multi-language & localization support
- Live sessions / webinar integration
- Corporate/team accounts (B2B LMS)

Phase 3 — Service Cost Breakdown

Service	Plan	Monthly Cost	Notes
All Phase 2 Services	Upgraded tiers	**~$220 / month**	Baseline carried forward from Phase 2
Algolia (Search)	Grow Plan	**$50–$100 / month**	Fast, scalable full-text search
AWS (Expanded Infra)	EC2 + Lambda + RDS	**$100–$300 / month**	Depends heavily on traffic & usage
Twilio / Daily.co (Live)	Pay per use	**$50–$200 / month**	For live sessions / webinars (optional)
Apple Dev Account	Annual	**~$8 / month**	$99/year for iOS app distribution
Google Play Account	One-time	**$0 (after $25 setup)**	$25 one-time fee for Android distribution
Monitoring (Datadog / Sentry)	Team Plan	**$50–$80 / month**	Error tracking, performance monitoring
PHASE 3 TOTAL (est.)		**$500–$900+ / month**	**Scales with platform traffic & users**

Note: Phase 3 costs are highly variable. Live sessions and AWS infrastructure scale directly with platform usage and active users. These are estimates based on a platform serving 500–2,000 active monthly users.

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5. Cost Summary — All Phases

The table below provides a high-level cost comparison across all three phases. These are infrastructure and third-party service costs only and exclude all development fees.

Phase	Timeline	Key Services	Monthly Cost	Status
Phase 1 — MVP	Month 1–3	Cloudinary, MongoDB, Render, SendGrid	**$0 – $8**	**Ready to Launch**
Phase 2 — Core LMS	Month 4–8	Mux/AWS, Stripe, MongoDB M10, SendGrid	**$155 – $220**	**Post-Validation**
Phase 3 — Scale	Month 9+	AWS, Algolia, Mobile, Live Sessions	**$500 – $900+**	**Growth Stage**

Key Financial Principles

1. **Start free, scale with revenue —** MVP costs nothing. Only upgrade services when paying users are onboarded.
2. **Stripe fees are not overhead —** Payment processing fees are deducted from revenue, not fixed costs.
3. **Phase 3 costs are usage-based —** Higher traffic = higher cost, but also higher revenue. Profitability improves with scale.
4. **Migration path is clean —** Services are designed to be swapped (e.g., Cloudinary → Mux) with minimal frontend changes.

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6. Technology Stack Reference

The following technologies are used to build the platform across all phases:

Layer	Technology	Purpose
Frontend	React.js + Vite + Tailwind CSS	Student, Instructor, and Admin UI
Backend	Node.js + Express.js	REST API, business logic, auth
Database	MongoDB + Mongoose	Users, courses, enrollments, transactions
Authentication	JWT + Google OAuth (Passport.js)	Secure user authentication
Video (MVP)	Cloudinary	Free video hosting and delivery
Video (Phase 2)	Mux or AWS S3 + CloudFront	HLS streaming, adaptive quality, DRM
Payments	Stripe + Stripe Connect	Course purchases, instructor payouts
Email	Nodemailer + SendGrid	Transactional emails, notifications
Search	MongoDB Text Index → Algolia (P3)	Course search and filtering
Hosting	Vercel (FE) + Render/Railway (BE)	Scalable cloud deployment

Disclaimer

All costs listed in this document are estimates based on publicly available pricing as of February 2026. Actual costs may vary based on usage, number of users, video hours consumed, and third-party pricing changes. Development costs, custom tooling, and internal team resources are excluded from this proposal.